

Maximize
 w, b

$$\frac{2}{\|w\|}$$

$\Rightarrow$

$$\min_{w, b} \frac{\|w\|}{2}$$

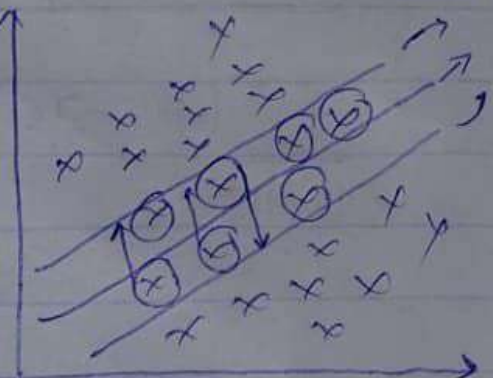
$$C_i = 6$$

cost function of svm (svc)

hyperparameters

$$\min_{w, b} \frac{\|w\|}{2} +$$

$$C \sum_{i=1}^n \xi_i$$



hinge loss

$\downarrow$
softmargin

allow many points we

want to avoid misclassification

$\rightarrow$ { summation of the distance of the incorrect data point from the marginal plane }